

PH 582: Assignment-1

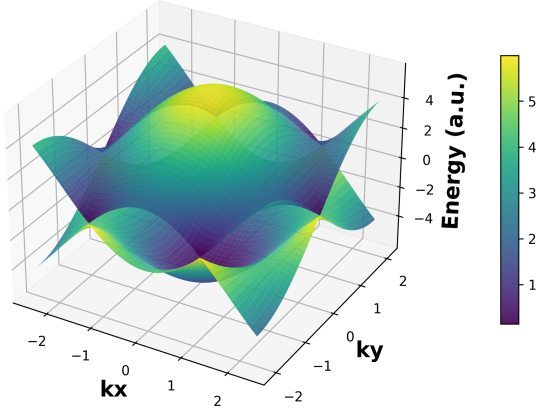
haldane Model Numerical Simulations

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Note: Parameters are taken in arbitrary units for numerical solution, e.g. nearest neighbor distance is taken as 1, and unless otherwise stated, t_1 is set to 1. The onsite energy shift is taken to be 0 but on-site energy offset is taken. If the question did not ask to plot for various parameters, a single set of parameters as stated in the report will have been used for the plot. Jupyter notebook was used for plotting using python and markdown cells can be used to find the code for any plot. Also the following short forms might be used at various places, both in report and the code:

- NN: Nearest neighbor (t_1)
 - NNN: Next nearest neighbor (t_2 with some phase)
 - BS: Band structure
 - M: Onsite energy offset
 - Ω : Berry curvature
 - $\mathcal{C}$: Chern number
 - TBD: Topological band diagram
 - ϕ : Phase for the complex NNN hopping parameter ($t_2 e^{i\phi}$)
 - IS: Inversion symmetry
 - TRS: Time reversal symmetry
1. **Band structure:** The band structure of both simple graphene and Haldane model are shown here. For graphene, parameters taken were: $t_1 = 1, t_2 = 0$. For Haldane model, $t_1 = 1, t_2 = 0.5, \phi = \pi/2, M = 0.3$. Band opening is seen in Haldane model with the parameters taken and the side-by-side comparison is shown in figure 1.
 2. **Berry curvature:** Berry curvature is plotted in figure 2. Parameters for the plots are in the caption. Two different plots are shown to show the change in simulated spatial variations upon changing NNN hopping.
 3. The topological band diagram is given in next question, so here I am showing the parameters for various important cases:

Band Structure (Graphene)



Band Structure (Haldane)

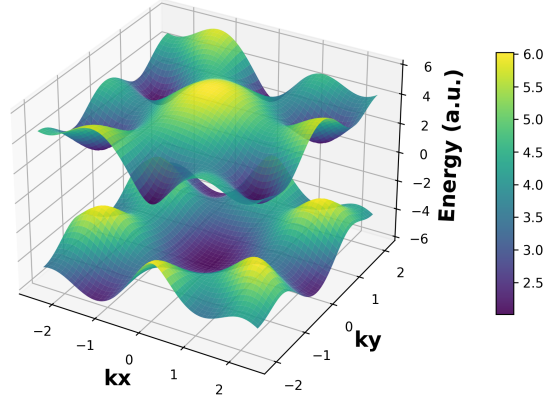


Figure 1: Band structures of Graphene (left) and Haldane (right). Color scale is correct for only upper bands in both plots, please refer to Energy axis labels for accuracy.

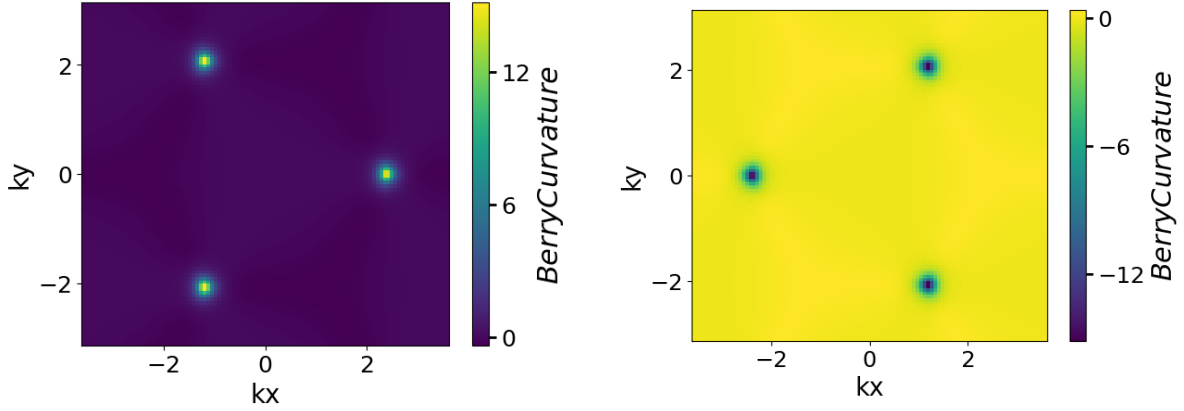


Figure 2: Berry curvature plot of Haldane model. Parameters sets used are: (left) $t_1 = 1, t_2 = -0.2, \phi = \pi/4, M = 1$ and (right) $t_1 = 1, t_2 = 0.2, \phi = \pi/4, M = 1$.

- Topological phase $\mathcal{C} = 1$: $t_1 = 1, t_2 = 0.3, \phi = \pi/4, M = 0.5$
 - Topological phase $\mathcal{C} = -1$: $t_1 = 1, t_2 = -0.3, \phi = \pi/4, M = 0.5$
 - Trivial phase $\mathcal{C} = 0$: $t_1 = 1, t_2 = 0, \phi = \pi/4, M = 0.5$
4. The topological band diagram is shown in figure 4 for two different sets of (t_1, t_2) , mentioned in caption.
 5. Chern number is not defined at the point of transition from one topological phase to another topological phase, which for Haldane model occurs at

$$M = \pm 3\sqrt{3}t_2 \sin \phi.$$

This is called a topological phase transition. Also, if $t_2 = 0$, then the bandgap can open at certain points for non-zero M , but there is no topological phase as there is no

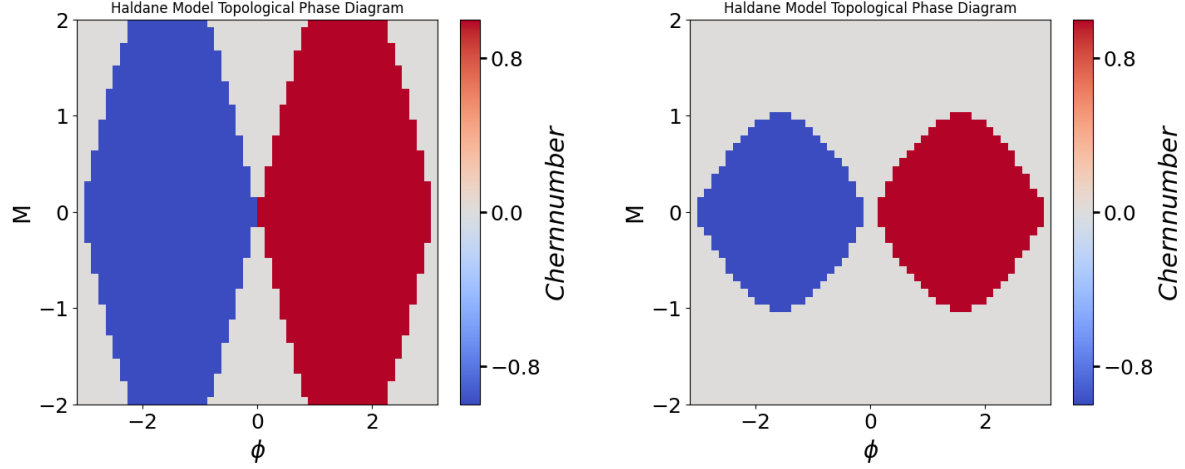


Figure 3: TBD plot of Haldane model. Parameters sets used are: (left) $t_1 = 1, t_2 = 0.5$ and (right) $t_1 = 1, t_2 = 0.2$. We only see integer Chern numbers since it is quantized. Chern number at $\phi = 0$ is not defined as it is the crossover from $\mathcal{C} = -1$ to $\mathcal{C} = +1$.

breaking of TRS. Only IS is broken if $M \neq 0$. For a defined non-zero Chern number, there can be no band crossings, else at such crossover points, there is said to be a topological phase transition and Chern number is not defined.